

HECTOR GENARO PACHECO

Software Engineer | Embedded Systems • IoT • Physical Systems

Southern California • hector.genaro.pacheco@gmail.com • Portfolio • [GitHub](#) • [LinkedIn](#)

PROFESSIONAL SUMMARY

Software Engineer combining computer science with a professional industrial Instrumentation & Controls background, experienced in both hardware systems engineering and collaborative software development. Track record includes embedded Linux systems on Raspberry Pi 5, programmatic camera control via gphoto2, real-time sensor telemetry pipelines, industrial calibration software, Jira/GitLab workflows, and systematic physical-world troubleshooting.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript, SQL, Bash	Embedded & Linux: Raspberry Pi 5, Debian/Linux, systemd Services, Kiosk Mode, GPIO, Active Cooling
Hardware Protocols: gphoto2, USB PTP, GPS/NMEA (UART/Serial), PIR Motion, 4–20 mA Loops, HART	Industrial Controls: PLC Systems & I/O, Relay Logic, 480VAC/120VAC/24VDC, Fluke 789, Trex HART
Networking & APIs: FastAPI, WebSockets, REST APIs, Node.js, Express, Docker	Software & Tools: Git, GitLab, Jira, React, PostgreSQL, Prisma ORM, Systematic Fault Isolation, Calibration Logs
Platforms: Palantir Foundry, Palantir AIP	Quality & Standards: Signal Validation, Preventative Maintenance, MPE Tolerances, Industrial Auditing

FEATURED SOFTWARE & SYSTEMS ENGINEERING PROJECTS

Interactive Photo Booth System — *Client Hardware/Software* **Raspberry Pi 5 • Linux (Debian) • Python • photobooth-app • gphoto2 • Canon DSLR • systemd Integration** [CASE STUDY ↗](#)

- Architected and built an event photo kiosk powered by an active-cooled Raspberry Pi 5 and 15.6" Acer touchscreen for a nonprofit client.
- Automated tethered Canon Rebel T100 DSLR control over USB PTP using gphoto2, managing camera state, shutter triggers, and photo transfers.
- Authored systemd user service definitions and kiosk boot scripts ensuring hands-off reboot recovery, persistent logging, and reliable event operation.

FieldTrack AI — *Edge Computing & Sensor Telemetry Platform* [GITHUB ↗](#) **Raspberry Pi 5 • Python • FastAPI • Node.js • React • WebSockets • GPS/NMEA**

- Built an edge telemetry platform on Raspberry Pi 5 integrating GPS/NMEA serial stream parsing, PIR motion hardware interrupts, and system diagnostics.
- Implemented Python/FastAPI hardware services streaming low-latency JSON telemetry frames over bidirectional WebSockets to a real-time React dashboard.

CalTrack — *Industrial Calibration SaaS & Mobile Platform* [LIVE APP ↗](#) **React • TypeScript • Node.js • Express • PostgreSQL • Docker • AWS ECS Fargate**

- [GITHUB ↗](#)
- Bridged physical industrial instrumentation workflows with modern cloud software, engineering an equipment calibration platform for process plant compliance.
 - Modeled database schemas for 4–20 mA transmitters (pressure, temperature, flow, level), tracking calibration histories, tolerances, and calibration cycles.

Additional Repositories & Deployments: **Pi Arcade OS** — Embedded Gaming Platform (Raspberry Pi/Python/GPIO/Pygame) [SITE ↗](#) [GITHUB ↗](#) • **EstateOS**

— Real Estate Intelligence (Deterministic Underwriting) [CASE STUDY ↗](#) • **Procurement Tracking** — Full-Stack GraphQL Workflow Platform [LIVE APP ↗](#)

[GITHUB ↗](#)

TECHNICAL & INDUSTRIAL EXPERIENCE

Software Engineering Contributor — *Nonprofit serving underserved communities* **Collaborative Contribution**

- Contributed to an existing codebase within a collaborative team environment, utilizing Jira issue tracking, Git/GitLab workflows, and peer technical communication.

Instrumentation & Controls Technician — *Process Control & Industrial Systems* **Field Engineering Experience**

- Calibrated and commissioned 4–20 mA process transmitters (pressure, temperature, flow, level) and troubleshoot PLC I/O wiring, relays, and industrial power loops with systematic fault isolation and auditable logs.

EDUCATION & PROFESSIONAL TRAINING

Western Governors University (WGU) — B.S. Software Engineering (In Progress, 2026–Present)
Mirion Technologies — SU-890 iCAM Operations & Maintenance (March 2026, 16.00 Continued Education Credits)
Palantir Technologies — Foundry & AIP Learning Credentials (May 2026)